

one query · 4 relevant passages exist in the corpus · top-10 returned

rank	1	2	3	4	5	6	7	8	9	10
relevant	.	●	.	.	●	.	.	.	●	.
$1/\log_2(i+1)$	1.000	0.631	0.500	0.431	0.387	0.356	0.333	0.315	0.301	0.289

$$\text{DCG@10} = 0.631 + 0.387 + 0.301 = \mathbf{1.3188}$$

$$\text{IDCG@10} = 1.000 + 0.631 + 0.500 + 0.431 = \mathbf{2.5616} \quad (\text{all 4 relevant at the top})$$

$$\mathbf{\text{Recall@10} = 0.75}$$

did we find them?
3 of 4 found

$$\mathbf{\text{MRR} = 0.50}$$

how soon was the first?
first hit at rank 2

$$\mathbf{\text{NDCG@10} = 0.5148}$$

are they near the top?
every hit discounted by rank

the same ranking scores 0.75, 0.50 and 0.515 — a document at rank 1 is worth 3.46 times one at rank 10, which is why only NDCG penalises burying a relevant passage